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THEMATIC ANALYSIS THEMES:

FOR INTRODUCTION

- **The Pedagogical Shift in the GenAI Era**
    - Generative AI is transforming higher education from a focus on banning the technology to integrating it as an active pedagogical partner. The goal is no longer just content delivery, but using AI to foster higher-order critical thinking.
- **"Productivity Paradox" & Cognitive Offloading**
    - Students increasingly use AI as a shortcut to maximize productivity and meet deadlines rather than to learn. This "cognitive offloading" allows them to generate final outputs while bypassing the mental friction required for actual memory retention.
- **Illusion of Truth and Confidence Miscalibration**
- Because AI generates text with a highly authoritative and fluent tone, students often blindly trust the output without verification. This creates a dangerous "illusion of competence," where a student's confidence in their knowledge is artificially high compared to their actual test scores.
- **Need for "Desirable Friction" (The Proposal Hook)**
    - To combat this false confidence, AI tools must be designed with "speed bumps" that force cognitive engagement rather than just handing out direct answers. This study proposes testing different AI interaction styles to see which type of friction best aligns a student's confidence with their true competence.

FOR SLR

- **Frameworks for Human-AI Collaboration**
    - Existing theoretical models (like the ICAP framework and Mollick's "Seven Ways") map human-AI interactions on a spectrum from passive consumption to interactive dialogue. These frameworks provide the theoretical justification for designing distinct, testable AI personas for this study.
- **The Prerequisite of AI Literacy & Foundational Knowledge**
    - AI is only an effective learning partner if students already possess a baseline understanding of the topic and the literacy to prompt the AI effectively. Without prior foundational knowledge, students cannot evaluate AI hallucinations, turning the tool back into a crutch.
- **The Passive Baseline (The "Crutch" Effect)**
    - When AI is used passively to provide direct answers (such as generating summaries or flashcards), it removes necessary learning friction. Literature shows this boosts immediate performance but ultimately harms long-term retention and skill transfer once the AI is removed.
- **Constructive Friction: The Socratic Tutor**
    - A Socratic AI introduces "constructive friction" by interrogating the student with open-ended questions and refusing to provide direct facts. This pedagogical method forces students to verbalize their reasoning, leading to significantly deeper conceptual understanding.
- **Generative Friction: Reverse Tutoring (The Protégé Effect)**
    - Reverse tutoring creates "generative friction" by having the student explain a concept to an AI acting as a confused novice. This forces students to organize their mental models and quickly exposes gaps in their own understanding, shattering the illusion of competence.
- **Psychological Impacts: Self-Efficacy, Trust, and Technostress**
    - Beyond just test scores, different AI interactions uniquely affect a student's emotional state, trust, and fatigue. Understanding these psychological impacts is crucial, as high-friction methods (like Socratic tutoring) might improve learning but also increase student frustration compared to human teachers.

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My Research Question:

Investigating the Impact of Generative AI Tutoring Styles on Student Knowledge Acquisition and Confidence Calibration

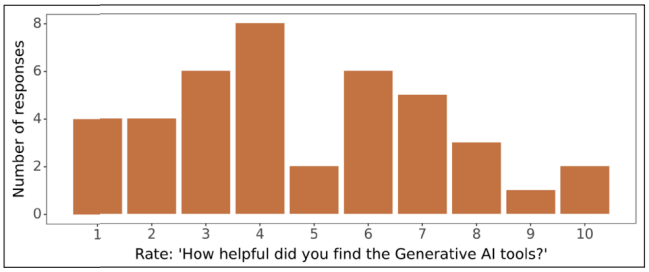
Literature Findings:

Approach Generative AI Tools Proactively or Risk Bypassing the Learning Process in Higher Education (Sallai et al., 2024)

Cons:
- Raises the concern of “death” of essays, which has risen from COVID because of increase in online take-home exams → students cheating by utilizing AI to write + paraphrase against plagiarism tools.
- Unreliable, inaccuracies in these tools
- Convincing tone of AI chatbots give illusion that output is true

Pros:
- GenAI can help non-native English speakers linguistically serve as teaching assistant
- GenAI minimizes pressure of homeworks/exams

Keynotes:
- Paper is about undergrad/postgrad students and how they use AI
    - ChatGPT used by almost all undergraduate & postgrad
        - Followed by Grammarly, then Co-pilot
    - They claim that it made it easier for them to learn at the start, but on the autumn term students have mixed perceptions



    -
    - Students in management find the tools less beneficial compared to quantitative courses (data sciences)
    - If Subject pace/delivery is easy to follow they less likely depend on AI
- Primary Analysis:

        - Students benefit most from GenAI when they understand the purpose and have basic concepts needed to complete the task.
            - AI is to help with the *meaningless* task
- Uni needs to create an environment where AI enhances human intelligence
- GenAI should be prevented if it removes opportunities to develop cognitive abilities/social skills
- Students that uses chatbots too much produce lower quality work → essays are generic
- Use of AI is not lazy → educators should realize this as an opportunity to boost student's interest
- Lack of minimal foundation knowledge makes it hard to validate Ai generated content → curriculum needs to be adjusted
- Abbas et al: excessive use of chatgpt reduces academic performance + memory
- Policy recommendation from the paper:
    - Assessment that requires documentation of the development to the final output, example: pitches/video updates.
    - Write down steps / strategies for students to help students draw connections
    - Delay adoption of Ai in learning new skill by aligning the foundation first, then AI becomes the assistant to help automate
    - Encourage student to track and share use of GenAI to support individual learning

The who, why, and how of ai-based chatbots for learning and teaching in higher education: A systematic review (Wenjuan et al.)

Utilizes active theory model → provides a comprehensive lens for examining learning processes (both outcomes but also facilitation of learning).
Wh?:
- undergraduates are more inclined to use AI chatbots

Why?
- Lack of **personalized learning experience**
- Lack of realism learning **interaction**
- Limitation of content/presentation
- Safe and controllable learning environment
    - Provide customized feedback, **flexibly adapt to student's needs**
    - Able to guide students to participate in **in-depth convos**

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Integrating Generative AI in University Teaching and Learning: A Model for Balanced Guidelines" (2024)

Pros:

- AI provides instant feedback + adaptive learning styles
- ChatGPT increases student engagement + satisfaction

Conts:

- Over-reliance & Procrastination, memory loss, decline in learning competence
- Plagiarizing in assignments

Proposed Balanced Approach:

- Instead of banning AI, Focus on decision making & training to enhance human capabilities → future readiness to enter Ai-driven industry
- Guidelines should be flexible to the context (faculty/departments' goal)
- Focus on honesty and ethical points, and fair use of AI such as citing if its from AI

Using AI to Implement Effective Teaching Strategies in Classrooms: Five Strategies, Including Prompts (Mollick & Mollick, 2023)

Strategies for educators on how AI can be effective:

- Provide example + explanations, analogies for abstract concepts
- Address student misconceptions → identify common misunderstandings
- Low Stake Testing → automate quizzes/tests creation
- Identify knowledge gap by assessing where students are struggling
- Help design materials that reinforce learning over time

Cautions

- Instructor/educate must use Ai as a tool, so it can evaluate AI hallucination
- Same prompt can product different result, try multiple times as needed
- It is to help "overwork instructors" by making complex pedagogy accessible

Assigning AI: Seven Ways of Using AI in Class (Mollick & Mollick, 2023)

- AI-Mentor (Providing Feedback): Acts as a guide to give students immediate, constructive feedback on their work.
- AI-Tutor (Direct Instruction): Provides personalized, step-by-step tutoring to help students master new concepts.
- AI-Coach (Increasing Metacognition): Prompts students to reflect on their learning process and think through problems.
- AI-Teammate (Collaborative Intelligence): Functions as a partner in group work, helping to brainstorm or provide diverse perspectives.
- AI-Student (Learning by Teaching): The student teaches the AI, which helps reinforce the student's own understanding.

- AI-Simulator (Practice Opportunities): Creates scenarios for students to practice skills in a safe, simulated environment.
- AI-Tool (Efficiency): Used for tasks like summarizing or outlining, though the authors warn this risk-averting role can lead to students missing out on developing core skills if overused.

Students must also be taught to verify AI outputs while educators. Educators also should create hard thinking tasks that AI can't replicate.

The CLEAR path: A framework for enhancing information literacy through prompt engineering (Lo, 2023)

CLEAR framework: a pedagogical tool designed to help users from lazy prompting to prompt engineering

- C: Concise, use direct instructions to keep Ai focused on core task
- L: Logical, information presented in step by step/naturally, AI will be more coherent
- E: Explicit, Tell the output you would look like (length, format, tone, points to include)
- A: Adaptive, if answer is general, pivot because prompting is iterative, keep refining
- R: reflective, users should analyze the response for accuracy and bias

Temparature:

Low 0.2 → focus on factual/boring answer

High 0.9 → creativity & word choices but increase risk of hallucinations

The paper argues that librarians should lead teachers to teach students how to engineer prompts to get better search results; however, humans in the loop are important because humans are responsible for the final output.

Exploring the effects of artificial intelligence on student and academic well-being in higher education (Klimova & Pikhart, 2025)

Ai has a dual impact on well being, it can be supportive but can be stressor as it can trigger fatigue or technostress.

Over-reliance on AI can also cause the student to reduce face2face interactions, declining the interpersonal skills and emotional intelligence → leading to social anxiety, isolation

It also blurs the line between leisure activities & academic task as it has become indispensable

Catalyst for future education: An empirical study on the Impact of artificial intelligence generated content on college students' innovation ability and autonomous learning (Wang et al., 2024)

Uses SEM (structural equation modeling) and meditation model, analyzes relationship between critical thinking, learning attitude, and the use of AI-generative content, innovation entrepreneurship abilities, and autonomous learning abilities.
Findings:

- Those with stronger critical thinking & positive learning attitude are more likely to use AI effectively
- Ai fosters both innovation & entrepreneurship abilities & autonomous learning
- The risk is in the decline of cognitive skills & concern regarding academic integrity.
    - Author argues that there is tension between student mastering AI & risk of over reliance (undermining independent thought)
- A central argument is use of AI should be integrated "judiciously" within the curriculum
- Educators should guide students in using AI prudently, discouraging over reliance and nurture critical thinking

Impact of Artificial Intelligence (AI) in Education on Students' Academic Development: Present and Future Prospects (Jivtode, 2024)

Ai has transformed teaching, improved student engagement by providing information based approach
Beyond pedagogy, Ai reduces administrative burden on educators, allowing more interaction with students.
Ai enables intelligent tutoring system that adapts to the student needs in real time, enhancing student engagement and making learning more immersive
Limitations of AI in education:

- Lack of human touch
- Data might be wrong → leading to false results
- Limits critical thinking and creativity development → over reliance → in the long term can limit students development (cognitive, social, emotional)
- Privacy concerns, ethical & bias issues
- AI can't guide like what teachers can do

Impacts of AI Usage Ethically on Students (Shamsi et al., 2024)

Adaptive learning systems improve academic results, however there are technical & ethical hurdles especially in contextual understandings in current Ai tools.

- AI must adhere to principle of fairness, transparency and responsibility
- Institution must provide guidelines to govern ethical use
- It should be a tool for deepening knowledge not shortcut.

Chatbots in education: A systematic review of objectives, underlying technology and theory, evaluation criteria, and impacts (Debets et al. 2025)

Primary use of chatbots in recent years has shifted from administrative assistance to pedagogical support,

- 65% of the reviewed chatbots are teaching oriented that focus on enhancing student knowledge & provide feedback
- 18% (20/71 papers) informed their chatbot is integrated with educational theory
- Author warns that adopting bots just because they are convenient not because they are aligned with learning goals
- Many papers lack theoretical explanations & foundations for the chatbot which may lead to misguided implementation

Integrating Generative AI in Cybersecurity Education: Case Study Insights on Pedagogical Strategies, Critical Thinking, and Responsible AI Use (Elkhodr & Gide, 2025)

- Argues that GenAI should transform students from informed receivers to critical evaluators → require students to critique AI → fostering higer order thinking skills rather than memorization
- Implemented a pedagogical model that consist of 2 distinct phases
    - Tutorial exercises → students refine AI outputs and identify the gaps in the AI's logic
    - Assesment Tasks → apply ai generate outputs to real world scenarios
- GenAI enhanced students ability to bridge theoretical knowledge to practical application, allowes generate risk assessments, enabling students to focus their cognitive effort on high level refinement
- AI must be use as instrucitonal scaffold supporting studnet, they should be thought the fundamental pedagogical strategy to verify AI's output
- GenAI most effective when embedded in real world scenarios, static AI use is discouraged, studnet must refine AI assisted outputs

To Tell or to Ask? Comparing the Effects of Targeted vs. Socratic AI Hints (Liding et al., 2026)

Compares Targeted hints (clear, concise suggestion) with Socratic hints (questions, analogies, examples). In the short term, Socratic performs worse in terms of efficiency, requiring more time, attempts and keystrokes. It also leads to higher error ratio (repeated errors 5%, and 14.7 in long term), they found no evidence that socratic hints improve long term outcomes or coding skills. Although having a higher level of difficulty, it reduces debugging efficiency. Despite the hints from Socratic being pedagogically sound, the targeted approach remained superior.

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A Comparative Study of Free Self-Explanations and Socratic Tutoring Explanations for Source Code Comprehension (Tamang et al., 2021),

Compares 2 instructional strategy in understanding programming code, free-elf explanation & socratic in learning java programming

- Socratic method achieved 59% learning gain. While the free self-explanation achieve 30% learning gain, outperforming by 29%
- Both works well regardless of prior knowledge (no significant difference)
- All 105 participant has prior knowledge of programming
    - Briefed about the goal
    - Given 6 source code example called **main task**,
    - They take post test about content in the pre-test and main taks for final score

From teachers to chatbots: Scaffolded corrective feedback and student trust in online L2 English classrooms (Soyoof et al., 2026)

investigates how scaffolded corrective feedback (SCF) provided by a human teacher compares to that provided by ChatGPT in terms of learning outcomes and student trust

- Student with human feedback outperform chatgpt both immediate & post test
- Qualitative Analysis shows Teacher has advantage in trust, student are able to trust teacher more than AI
- Ai feedback are found to be mechanical, repetitive, and boring, and generation sto[[ong frustrated student & undermined their trust
- ChatGPT feedback are one directional and static due to predefined prompts
- AI should be used by teachers in the loop to make it meaningful
- Grounded in Vygotsky's sociocultural theory, the paper argues that for feedback to be effective, it must be collaborative and situated within the learner's **Zone of Proximal Development (ZPD)**. Teachers were better at identifying and staying within this zone than the generic AI.

The test is conducted with 40 learners of EnglishForeignLanguage, test is for accurate use of english article (a, an, the) → one of most frequent & persistent errors made by L2,

- Pre test given, 2 post test (1 week after (short term effect), and 1 month after (durability of knowledge))
- 20 fill in the blank items where students had to supply the correct article
- Thematic analysis with semi structured interviews

GELEX: Generative AI-Hybrid System for Example-Based Learning (Yazici et al., 2024)

Novel system combining LLM procedural with constraints to improve student learning from examples.
Example based learning are usually static, can't adapt to students misconception. GELEX uses hybrid architecture where LLM generates content, but the an "expert module" efncores pedagogical constraints to ensure the examples are logical

- Use of Prompt engineer for pedagogy with **guardrails can force LLM to act as socratic tutor**
- Reduces workload, because expert can just give base example and AI can build dozens of variation based on students needs
- Tested for logic and procedural tasks

Putting Things into Context: Generative AI-Enabled Context Personalization for Vocabulary Learning Improves Learning Motivation (Leong et al., 2024)

Explores how LLM can be use to personalize contextual sentences into vocabulary learning to match students individual interests

- Personalizing learning context can improve motivation and engagement compared to using generic non personalized examples
- Can incorporate vocabulary into a chosen interest such as cooking/scifi/sports
- Students are more likely to recall and apply new words when the word is used in topics that have interest in
- Make sure the "interesting" context does not distract the learner
- Compares personalized based on user profile, and generic non personalized, uses diverse group of learners to improve their english proficiency
- Measurements: Motivation (IMMS), Learning Outcome (Post test using word definition recall and usage), Qualitative feedback with semi-structured interviews on naturalness and helpfulness of the AI-generated sentences

The authors conclude that LLM-enabled personalization is a transformative tool for language education, offering a way to make traditionally "dry" subjects like vocabulary building more immersive and personally relevant.

Generative AI Based on Deep Knowledge Tracing for Academic Insight and Intelligent Learning Strategy Optimization (Wang, 2026)

Explores integration of deep learning models with gen AI to create prices & adaptive education environments.
→ model can track students degree of mastery in real time, allows educators to identify specific knowledge gaps
→ optimizes student learning strategies, AI can recommend resources or paths tailored to what the student hasn't mastered yet.

Socratic AI Against Disinformation: Improving Critical Thinking to Recognize Disinformation Using Socratic AI (Duelen et al., 2024)

Development of an AI driven tool designed to bolster resistance to fak news by using guiding questions:

- Simply flagging content as fake is ineffective, users need to develop internal critical thinking to evaluate themselves
- Uses a series of structured questions based on the ***Island approach*** to guide user through a critical evaluation of th source
    - Island approach: a qualitative projective technique to evaluate large number of attributes, features, or scenarios
- **Socratic method helps users recognize their own cognitive biases & emotional triggers**. It also empower users to be more cognitively active compared to passive consumption of fact checking

Enhancing Adaptive Personalized Learning Interfaces with Generative AI for Individuals with ADHD (Gunawardana et al., 2025)

Gen Ai to be used as ADHD-friendly user interfaces for LMS

- Traditional LMS contains cognitive noise (distraction, complex layouts, vague instruction)
- Gen AI can act as accessibility architect by using specific accessibility focused personas and constraints in prompts making it adhd friendly
    - Minimalism, explicit focus state and standardized colro palettes
    - Students can use genAI to adjust the interface according to their needs
    - Leads to a more effective and empathetic interface design
- Gen AI is revolutionary for adaptive personalized learning

Generative AI and Gamification for Personalized Learning: Literature Review and Future Challenges (Abbes et al., 2024)

Examines genAI & gamification to create highly adaptive & engaging educational environment

- Author argues that gamification adjusted to user personality, value & motivation can make the learning process mor fun and rigorous
- GAI can be the cornerstone of **education 5.0**, prioritizes personalized and adaptive learning experience over standardized instruction
    - Education 4.0 -> alignment of education practices with 4th industrial revolution, use of big data, iOT, and AI to support learning preparing students for highly digitalized workforce
    - Education 5.0 -> Human-centric personalization, individualized & adaptive based on specific personality, values and motivations. Encourages self assessment and empower students to take ownership of their own learning journey using AI driven tools.
- Practical examples can be seen in Quizizz AI (Wayground) for self assessment allowing student tot take ownership of their learning journey

- However currently the field lacks models that are able to dynamically adapt the content generate & engagement strategy to ensure AI driven environments are motivation and effective.

Generative AI without guardrails can harm learning: Evidence from high school mathematics (Bastani et al., 2025)

- Explores design of AI tutoring tools impact on performance in math
- Shows boost in practice, but leads to deficit when tasked to perform tasks with no AI assistance
- Uses GPT Bases → direct answer, and GPT Tutor → socratic tutor with hints & guidance
    - Base → 48% improvement in practice, Tutor 127% improvement in practice
    - When Ai access removed, GPT base perform 17% worse than no AI control group
    - GPT Tutor did not show learning loss, equal to control group
    - AI can act as a crutch, allows student to complete task without acquiring underlying skills

Study structure:

- 955 students -> 3 groups → base, tutor, control,
- 3 phase → practice (using tools), in-tool testing (measure performance while tools active), transfer phase: student tested without AI with similar problems
- Uses Mathematics topics, ensuring the results were relevant to standard education
- GenAI can be a double edged sword in education, can improve productivity & immediate performance, but lacks guardrails and risks turning into answer machines. Emphasize that design of AI interaction is more important for learning than the model capabilities itself.

Integrating Generative AI with Human-Centered Pedagogy: An Innovative Path for Vocational Education" (Xu & Chen, 2026)

Integration of genAI into vocational training,

- Author argues that genAI shouldn't just be a tool for content delivery but a partner in human centered pedagogy that focuses ont eh students creative and critical thinking
- Case based evidence where AI was successfully integrated:
    - Art & design → image generation
    - Computer Science → RAG that provides context ware answers from notes
    - Math → automated analysis reports for teacher to gain insight
- Human must remain the final judge of quality and relevance, Ai to be used as digital whiteboard
- GenAI allows students to focus on higher order skills such as synthesis, critique, innovative design, and have genAI provide creative scaffolding and automating repetitive tasks

Students-Generative AI interaction patterns and its impact on academic writing" (Kim et al., 2026)

Students with high AI literacy adopted a collaborative approach, engaging AI in planning & evaluation, while low AI literacy is less interactive choosing to ideate & evaluate independently and uses AI in a linear fashion
Student with high literacy outperform across all evaluation rubric, content structure/organization & expression
High literacy takes AI as a partner rather than a mere tool, and low literacy over-rely on AI generated content without critical assessment → poor long term learning outcome.
Instructors should focus on making students flexible cognitively, and guide them to use AI for high order reasoning rather than just tools for content generation.

Evolution of Learning: Assessing the Transformative Impact of Generative AI on Higher Education" (Krause et al., 2025)

Students primarily utilize GenAI for practical, task-oriented activities like assignment writing and exam prep (creating summaries)
GenAi isn't just a tool but an emerging tech that is changing how knowledge is accessed and processed in academic settings.
Institutions must move beyond general policies to develop guidelines that address the unique needs faced by both students & teachers.

The impact of Generative AI on students' learning: a study of learning satisfaction, self-efficacy and learning outcomes" (Yang et al., 2025)

Compare effectiveness of gen ai assisted learning compared to traditional teacher for college students. Students achieve higher learning outcomes when (test scores) when assistant in gen AI compared to teachers.
Students in the teacher group has higher self-efficacy than the AI group, → AI helps students perform better, but teachers are effective in building student confidence in their own abilities.
No significant difference in learning satisfaction, both are satisfied,
AI can be powerful for enhancing task performance & completion, but it can't replicate psychological or motivational support provided by human instructors.
Metrics measured: Learning satisfaction, self-efficacy, and learning outcome using a post-test

Supporting Teaching and Learning Computational Thinking Skills with Generative AI in Computing Education" (Benedetti, 2025)

CT is important skill for achieving goal in computation driven world, beyond just how to code, GenAI can support CT dimensions such as abstraction, decomposition, algorithmic thinking & pattern recognition, Gen Ai is powerful ally in teaching computational thinking, focusing on the process of thinking & problem solving rather than just the final code output, allows student to develop robust CT skill that can be use for various scientific & professional domains.

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Pedagogical Applications of Generative AI in Higher Education: A Systematic Review of the Field (Qian, 2025)

How higher education adapted to genAI during 2 years
→ focus towards using genAI to forst higher-order skills → creativity, critical thinking, learning autonomy & prompt literacy
→ AI enhances productivity, but risk students outsourcing critical cognitive tasks to the model (brain drain)
→ prompting has become a foundational academic competency, essential for students to effectively steer AI towards a more educational outcome
→ future research and practice must focus on exploratory, human-centric models ensuring AI serves as a cognitive partner rather than cognitive replacement.

Bridging pedagogy and technology: a generative AI and IoT approach to transformative English language education (Li, 2025)
→ explores development of AI-driven oral assessment tool for english as a second language education
→ G-ASR tool precision accuracy of 94,7% on native speaker, 86.6% on non-native speaker
→ more interactive than teacher, Ai 60%, teacher 40%

Chatting with a Learning Analytics Dashboard: The Role of Generative AI Literacy on Learner Interaction with Conventional and Scaffolding Chatbots" (Jin et al., 2025)

GenAI literacy effect with different types of Ai chatbots integrated into learning analytics dashboard
→ students with higher genAI literacy are better at steering the AI to provide deeper insights into their own learning pattern.
→ compares 2 Ai chatbots:

- Conventional → direct answers
- Scaffolding → guide using follow up questions → asking deeper analysis & self-reflection

Learners with low literacy struggle with scaffolding bots, finding it frustrating and often reverting to simple transactional queries
GenAi can become an interpreter between the raw data to student's learning goal
Study Strategy:

- Participants were assessed on their GenAI literacy → ability to understand, use and evaluate genAi models before experiment
- Randomized assignment of conventional/scaffolding
- Analysis from: Chatlogs, Dashboard interactions, self-reported surveys

GenAI is essential for productive human-AI collaboration in learning, but it shouldn't just provide answers, it should scaffold students'own critical thinking process, provide the student the foundational AI literacy to engage with the system.

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The Impact of Generative AI on Postgraduate Study (Wang et al., 2026)

How postgraduate utilize genAI and the influence on their research and academic tasks

- Postgrad use genAi for high level research support → literature synthesis & data interpretation
- Productivity Paradox, Ai speeds up drafting but lead to reduction in the depth of critical analysis
- Higher awareness of academic integrity & risk of AI hallucination
- Study strategy uses AI usage log → documenting instances they interacted with the AI, then semi-structured interviews to explore the changing relationship
  Requires humans to have a healthy skepticism or oversight of all the AI-generated contributions.

Navigating the New Information Landscape: A Study of Generative AI-Powered Search Engines in Higher Education (Miller & Thompson, 2026)
→ AI driven search tools are transforming student research habits
→ moving away from list based search result to AI summaries that synthesizes information from multiple sources
→ Student prioritizes speed & convenience of AI summaries over rigorous verification of underlying sources
→ Students are proficient at prompting, but lack the skills to evaluate synthetic coherence of Ai response → overlooking hallucination
Study Strategy:
→ 85 students are tasked to research on a multi-sided social-technical issue (ethics of deep sea-mining), splitted into 2 groups of traditional search engine and AI powered, uses Eye tracking to see which part of AI summaries students focused on, post task eval to rate the credibility of claims and trust, reflection interviews to see how their mental model works

Generative AI to support self-regulated learning: a systematic review of the literature (Lodge et al., 2026)

→ same as previous paper, the risk of cognitive outsourcing, when student rely heavily on AI to plan organize and evaluate their work, student dail to develop internal self regulation skills
→ GenAi arent yet supportive of self regulated learning, they must be pedagogically engineered to provide hints & scaffold rather than direct answers

Beyond the Hype: A Comprehensive Review of Current Trends in Generative AI Research, Teaching Practices, and Tools (Prather et al., 2025)

→ Ai also shifting the skills required by software dev, from syntax heavy coding to higher level architectural design & AI augmented problem solving
→ there is a trend from banning genAI towards integrating it into teaching practices by using hybrid models that combine traditional pedagogy with Ai assisted exploration

→ prompt engineering is not just a technical trick but a required competency in future computing professionals
→ GenAI are Lowering the entry barrier for complex computing tasks, student can engage with advanced concepts
→ AI can generate plausible outputs, therefore human oversight is needed

Large language models for education: An open-source paradigm for automated Q&A in the graduate classroom (Perez et al., 2026)

→ Ai driven pipeline gave effective feedback same as teaching assistants in terms of accuracy & specificity
→ the effectiveness of the RAG in mitigating hallucinations to ensure it is grounded just by the courses provided materials
→ this open source paradigm can provide high quality education and overcome cost & accessibility barrier
→ successfully provide scaffolding → give guides than just direct answers
→ responses from AI compared against human experts through blinded review where evaluator doesn't know the source of the response.

Measuring undergraduate students' reliance on Generative AI during problem-solving: Scale development and validation (Hou et al., 2025)

Study specifies patterns of reliance such as:
→ Initial consultation → using Ai to kickstart brainstorming or decomposition of problem
→ Verification seeking → dependent on AI to check the logic/accuracy of their the student's generated solution
→ Content delegation → adopting Ai generated content into their final output
Student who is blindly reliance on AI experience lower cognitive engagement, while those who uses Ai as critical partner showed improved problem solving strategies

Retrieval Practice Consistently Benefits Student Learning: a Systematic Review of Applied Research in Schools and Classrooms" (Agarwal et al., 2021)

Retrieval learning is where students recall/recognize info from their previously learned in order to strengthen memory and comprehension.
Retrieval practice is one of the most reliable strategy → out of 50 experiments, 57% revealed medium to large benefits of student learning
It is powerful but can be enhanced by providing feedback after retrieval attempt to enhance learning gains & correct misconceptions.

Preparing Students for an AI-Driven World: Generative AI and Curriculum Reform in Higher Education (Ma et al., 2025)

Argues that GenAI is no longer optional curriculum but a necessity in higher education survival due to the rapid professional landscape.
Ai literacy can be put into 3 tier:
→ Fundamental concept: Understand basic logic and ethics of AI
→ Applied use: mastering domain-specific AI tools to enhance productivity
→ Advanced technique: develop the ability to create or customize AI workflow

- The major finding of the paper is the pedagogical shift from memorization to problem solving & active learning strategies, such as project-based or interdisciplinary collaboration.
- Human-centric competencies also remain important such as critical judgment, ethical judgement

By adopting a strategy that is modular, tiered, and problem-focused, institutions can ensure GenAI serves as a catalyst for innovation rather than academic integrity thread.

Scaffolding Learning Scenarios with a Socratic Chatbot: Insights from Educators (Hilliger et al., 2026)

Investigates adoption of instructional design of Socratic chatbot to support student self reflection
Socrative chatbot serves 2 roles:

- Provide personalized support for students
- Assist teachers with instructional design to rethink teaching methods

85% of educators agreed that the AI Chatbot is useful for promoting self-regulated learning and is easy to learn and use.
The socratic chatbot responds with open-ended questions & reflective prompts rather than providing direct answers, which fosters critical thinking.
Built using Gpt-4o-mini, conducted with 44 higher education researchers & practitioners, given a demo of the tool & then tasked with proposing their own learning scenarios.
Collected data from 20 participants through online survey with 5 likert scale to measure usability and usefulness.
8 learning scenarios primarily in computer science, language learning, mathematics, which were analyzed to understand how educators envision integrating the bot into their workflows.

Boosting Large Language Models with Socratic Method for Conversational Mathematics Teaching (Ding et al., 2024)

Framework to transform LLMs into answer engines for Socratic tutors, in mathematics education.

- Identified that standard LLM typically provide direct solution to math problems,which deprive students of the cognitive effort
- The back & forth dialogue of the socratic chatbot encourages students to verbalize their reasoning process, identifying and correcting underlying misconceptions.

- Compare chain of thought and tree of thought to see which is better, models are not just evaluated on the correctness of the final answer but also the teaching quality.
- The methodology involved analyzing long-context dialogues, interaction turns without losing the context of the student's specific struggle.

SocratiQ: A Generative AI-Powered Learning Companion for Personalized Education and Broader Accessibility (Jabbour et al., 2025)

Socratiq → provide benefits of 1on1 socratic tutoring, which is traditionally resource-intensive human process to students at scale through genAI
The platform is engineered to prevent the "direct answer" trap, uses a conversational engine that guides students through the complex topics.

Teaching and Learning in the Age of Generative AI: Evidence-Based Approaches to Pedagogy, Ethics, and Beyond (Pham, 2026)

Emphasizes the critical need for educators to reflect on ethical implications of using ChatGPT, dall-e and gemini in educational settings.
Global standards to define competencies required for the AI era:

- UNESCO AI Competency Framework a structured approach to what students & educators need to know do and understand regarding AI

Transforming Education: The Impact of Generative AI

While GenAI enables increased productivity, it poses a dilemma due tot he lack of established rules & moral standards in education
A survey shows that 100% of sampled students support ChatGPT, but they are concerned on the authenticity & reliability of the content generated.

he Use of Generative AI in Education: Opportunities, Challenges, and Future Directions" (Alshumaimeri et al., 2026)

The ability of Ai to generate high quality essays and reports necessitates a shift in assessment towards the LEARNING PROCESS, such as incorporating oral examination and critical AI-output evaluations.
Highlights the daunting challenges such as algorithmic bias, data privacy concern and also potential depersonalization in the teach-student relationship.
AI can assist,- but teacher autonomy is vital to be preserved to ensure that pedagogical decision remain human-centric and ethically grounded.

Active Methods to possible try:

In the design of the study, we can compare up to 3 active strategies studying with AI, and have a passive control. Here are the selection of active strategies that we can choose from:

Socratic (WILL USE)

AI asks open-ended questions, where students justify their answer while connecting concepts. Students need to learn the direct facts (OpenAI's study mode).
ICAP framework (Chi & Wylie, 2014):

- Interactive → highest level, : dialogue & join construction of knowledge
- Constructive, student generate new ideas beyond learning materials (self explaining)
- active, → physical manip of focus attention taking notes, highlighting
- passive → lowest level : receive info without performing any action

Fakour & Imani 2025:

- Socratic questioning → method of asking thought provoking question than direct answers stimulating student reflection
- Critical Thinking Dimensions measured:
    - Analysis: break down complex infos
    - Evaluation: Assessing credibility / logical strength of argument
    - Inference: Logical conclusion from data
    - Self-Regulation: monitoring & correcting one's own cognitive process
- AI shouldn't be to replace humans but as a learning process fostering intellectual humility and persistent questioning.

Favero et al., 2024

- Bot uses fine tuning to ask structured thought provoking questions
- Uses smaller openSource models Llama-2 7B & 13B, address concerns on data privacy and democratized access to high quality education tools
- Ai finds when student is stuck and provide scaffolding → guides to student conclusion
- The bot can avoid repetition trap & is good at staying on topic
- Socratic method like this is good for theoretical concepts

In terms of for learning how to code, based on (*Alshaik et al., 2020*) socratic guides student to reason through the logic, key concepts of source code and how it functions, → students showed better improvement in programming concepts compared to their pre-test scores, breaks down complex code into logical segments that require cognitive processing

Reverse Tutoring (WILL USE)

Student is the teacher and explain concepts to the AI, the AI act like a 10 year old that asks clarifying questions ("Wait, but why does X happen")

- Okita & Schwarts, 2013 explore learning by teaching using a mechanism called Recursive Feedback, the tutor observe how their students apply what's been taught
- Protege effects → people learn harder when they know they must teach, significant learning gain occur after the teaching sessions → tutor observe the student's outcome

- Observing the students mistake/successes allows tutor to make inference about their own understanding, if student fails the tutor's own explanation may have been incomplete or flawed → triggering self correction
    - They do propose that having a 3rd party is important so that when teaching to a new person, the knowledge must be robust enough to be generalize

Scaffolder

AI gives problems to solve, student answers, if they get it wrong, AI dont give answer but hints, hint keeps getting larger the more the mistake
Shabani et al, 2010
Application of Vygotsky Zone of Proximal Development → distance between learner development level and their potential development (what they can do alone vs with guidance)

- Learning is a social process, traditional test only measure their past development, in contrast dynamic assessment interact within their ZPD to reveal future potential
- Instruction should be designed to encourage peer interaction so they can scaffold each other

Cognitive Conflict | Debate

Student explains a concept, AI debates and takes the incorrect stance, arguing with the student, the student must defend their knowledge & point out AI's mistakes
(Singer, 2018)
Study is using **nature of science**

- Combining cognitive conflict success is limited by how students perceive their owb=n errors, educators are encouraged to use targeted conceptual questions that force students to confront & resolve their internal logic.

(Limon, 2001(https://doi.org/10.1016/S0959-4752(00)00037-2))

- A bit difficult, it doesn't guarantee
- Student overlooks info that doesn't fit their mental model, might even twist new evidence to support their existing theory
- To make it work, student must have high metacognitive awareness, monitor own understanding, other than that student must also have:
    - Prior Knowledge, and motivation to engage in the debates

(Sumadji & Yuwono, 2020)

- 3 stages of resolution
    - Present a problem outside of the student logic
    - Make sure student recognize their answer is inconsistent
    - Guide the student to reorganize their cognitive structure

Developing cognitive conflict is highly effective, though complex, for math, but requires teachers to have deep understanding of common misconceptions.

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Erroneous Example | Error-Driven Learning

AI gives summarization recaps that has deliberate errors, and student must find the mistakes and correct the AI's mistakes

Finding from the paper (Tsovaltzi et al., 2010)

- Benefits **vary by grade level,** higher grade gain cognitive and conceptually, lower grade showed metacognitive learning gains, while middle grade showed no significant benefit
- Must have the role of **scaffolding to assist in detecting and correcting errors** (learner must **have prior knowledge**)
- Self-explaining both correct & incorrect examples strengthen correct strategies & weakening incorrect ones
- Mathematics teachers remain skeptical of the practice (student might assimilate incorrect points)

ZONGS (TO BE EXCLUDED):

When to Ask a Question: Understanding Communication Strategies in Generative AI Tools (Park et al., 2025)
The authors conclude that the method of obtaining user information is just as essential as the information itself. Developers should not rely solely on the "chatbot" iterative model; instead, they should leverage distributional data to actively determine the optimal interaction strategy based on whether they are prioritizing efficiency, fairness, or a balance of both.